

Face Recognition-Based Mass Attendance Using YOLOv5 and ArcFace

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Abstract. Manually recording attendance of a mass gathering like a classroom or seminar is very difficult and error-prone. To make the attendance process trouble-free, in this paper, we have explored the application of face recognition in the automated attendance system that can be used for educational and organizational purposes. Face recognition is a computer vision technology that can be applied for different purposes in various practical applications. The first task of face recognition is to detect faces from an image or real-time video input. Different state-of-the-art Deep Neural Network-based face detection and face recognition models are available in computer vision. We have studied popular face detection models like HaarCascade, SSD, MTCNN, and YOLOv5. We have also analyzed LBPH, FaceNet, ArcFace, and DeepFace models for the face recognition tasks. For this study, we have collected an image dataset of faces using a real-life classroom environment. Based on the result of our study we have proposed an automated attendance system using the YOLOv5 and ArcFace model for mass attendance, that is, recording the attendance of all persons in a room at a time.

Keywords: Face recognition · Automated attendance system · Deep Learning · CNN

1 Introduction

Human assembly (e.g., meetings, rallies, seminars, workshops, classes) is a regular and crucial activity in our personal and professional life. In such gatherings, it is also important and sometimes necessary to identify or recognize present and absent persons in the program. In an educational institution, recording attendance in the learning management system is crucial for students in in-person or online classes. Attendance carries equal importance for teachers and other officials when they join in any academic or departmental meeting. We can categorize the process of collecting attendance information into two classes: manual and automated. If we consider a classroom scenario, in the manual attendance process teacher calls the individual student's roll number or name sequentially, and based on the student's response, he/she marks the present or absent status

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of the student on a paper sheet. This manual attendance process is undoubtedly time-consuming, laborious, and, importantly, error-prone. Because one student can imitate another student's attendance, and the teacher can also make errors when marking on a paper sheet. The automated attendance process can resolve some of these problems. Several methods can be equipped to record attendance in the automated process, such as authentication by user ID and password, smart ID card, GPS tracking, and fingerprint biometric devices. The automated attendance system leverages physical labor and reduces time by recording students' attendance using technology without human involvement. However, it also has another issue of false attendance because one student can bring other students smart ID cards who were absent in class. Also, they can share this information with other students if we use authentication methods through user ID and password. To solve these problems of attendance systems, we can apply the latest computer vision trend of the facial recognition system for students' attendance (Fig. 1).

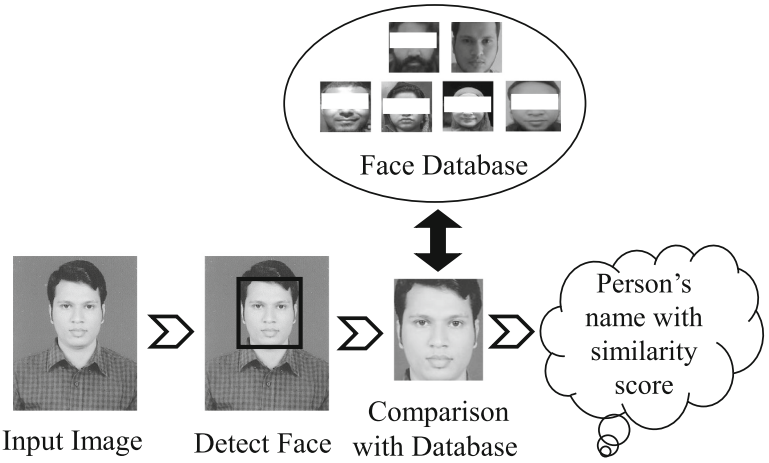


Fig. 1. Generic Face Recognition Process

Face recognition is a common human phenomenon that we unconsciously use every time we meet with other people to recognize the person's identity. Formally face recognition is the process of finding the identity of any person from an image or video automatically by using computer technology. A face recognition system constructs a database containing facial images of all known persons and the features of these images. When an image or video is provided, the recognition system compares the features of the input facial image with known images and features stored in the database, and the person with the maximum similarity value is considered the identified person for that input image. If the similarity is less than some threshold for all people in the database, it is identified as an unknown face. This technique can be easily used in a classroom environment by installing a high-quality camera from which all student's faces can be visible. Using face recognition in the classroom, we can record attendance and monitor

students' activities in class and exam hall. We need to consider several factors to implement face recognition for attendance in the classroom environment.

Face recognition systems need to consider the changes in facial expression, appearance, hairstyle, facial hair, etc. Also, lighting, scaling factor, distance, face angle, occlusion, etc. make face recognition a challenging problem in computer vision and image processing. Therefore, a face recognition system has to be robust and accurate enough to tackle all these factors. Many traditional approaches are available in computer vision to recognize faces from image or video, like Laplacian of Gaussian (LoG), Principal component analysis (PCA) [18], Independent Component Analysis (ICA) [21], Eigenface [20], Histogram oriented gradient (HoG) [10], Local Binary Pattern Histogram (LBPH) [11,12], etc. These systems suffer badly in recognition accuracy when changes happen in light, brightness, angle, scale, etc., of the input images with the features stored in the database. In recent years face recognition has come to the attention of researchers as machine learning algorithms have been improved a lot in calculation due to the advancement in GPU and memory. Different authors applied machine learning algorithms like Support Vector Machine (SVM), Linear Discriminant Analysis (LDA) [2], Bayes classifier, AdaBoost [19], etc. for facial recognition tasks through image processing. After the development of Deep Learning Networks. Different authors concentrated on applying this technology in face recognition to enhance the accuracy and robustness of recognition systems and significantly contributed to improving results for different problems. Convolutional Neural Network (CNN), AlexNet CNN, FaceNet [14], Mobile FaceNet [13], ArcFace [17], OpenFace [22], DeepFace [16], YOLOv5 [8], Ultra-light Face [23], etc. algorithms from deep learning family can also be used in face recognition system.

This paper represents a real-time attendance system using Deep Learning-based face recognition in a real-life environmental setup. A robust dataset was created using the classroom setup to train the face recognition model that contains images of different angles, brightness, scales, etc. Pretrained models were used in this study to reduce the overfitting of the face recognition models. We have compared different algorithms using the same image dataset. Some related previous works are discussed in Sect. 2. Section 3 discusses different face detection and face recognition models used in this experiment. Section 4 contains system details, and Sect. 5 represents the dataset collection process. In Sect. 6, we have discussed how we have trained these models, and in Sect. 7, we have stated our experimental setup and evaluation results of these models. Finally, the conclusion and future work of this paper is mentioned in Sect. 8.

2 Related Works

Charan et al. in [28] proposed a face recognition-based attendance system to record students' attendance from the video of the classroom. The authors used the Haar-Cascade classifier to detect faces from video and the Local Binary Pattern Histogram algorithm for the identification of the detected face identity. From the OpenCV library, authors have used the detecMultiScale function to detect faces of different sizes.

In [29], Alhanaee et al. implemented three different face recognition models for a smart attendance system. Here the authors collected a dataset of 10 persons and used pertained models of AlexNet, GoogleNet, and SqueezeNet on this dataset. The data augmentation method was used to increase the number of training and testing images for each person. Within these three models, the accuracy of the AlexNet model is better on their dataset.

Another face recognition-based attendance system using Convolutional Neural Network and K-NN classifier was developed by Sutabri et al. [30]. This web-based system captures students' facial images using a Raspberry Pi camera and transmits these images to the image processing system. Then using Dlib's CNN model the system detects the faces from the input images and generates a 128-D embedding. Finally, the K-NN classifier was applied to this embedding to generate the face identity. This system can recognize one student at a time.

An attendance system based on the YOLOv3 model was proposed in the study of [31]. The YOLOv3 is a deep convolutional neural network, which uses darknet53 as the backbone of the network. According to the study this model was used for both the task of face detection and face recognition. The authors trained and tested this model on 19 person's faces. During the class period, each student needs to capture a face image using a webcam, then the system will recognize the face and record the attendance of the student.

All of these face recognition-based attendance systems use the convolutional neural network. But these systems are not suitable for recording the mass attendance of all the students in a classroom at a glance. Also, these systems were not trained and tested on a large number of students. So in this paper, we have proposed a face recognition-based attendance system using the YOLOv5 and ArcFace face recognition models. This system can recognize a large number of students simultaneously.

3 Methodology

A large number of state-of-the-art deep learning-based models are available in the field of image processing, those can be used for facial image analysis. These models can be categorized broadly into two classes, face detection models and face recognition models. For developing an automated attendance system first we have to apply a face detection algorithm to extract faces from live video input, then using a face recognition algorithm we can ensure the identity of the face image. This section will discuss some state-of-the-art algorithms used for face detection and recognition.

3.1 Face Detection Algorithms

Face detection plays a vital role in different computer vision-based applications. To perform any processing or apply any method on human faces available in the video or image, we first need to extract faces from that input. Face detection is a challenging issue in real-time applications due to the variation of illumination,

face position, background, movement, etc. However, many established and well-renowned algorithms are available for face detection from the input image and video.

A very typical face detection algorithm was introduced by Viola and Jones [1–3], which is called the HaarCascade face detection algorithm. The authors used Harr-like features in a machine learning-based method in this algorithm to detect faces. There are around 16000+ features in this algorithm, so the authors used the AdaBoost algorithm to select the best features to detect human faces. According to authors [2,3] if 200 features are used, the algorithm can achieve 95% accuracy in face detection, and finally, they suggested that 6000 features are enough to detect faces successfully.

Single Shot MultiBox Detector (SSD) is a DNN-based face object detection model that uses only one forward pass of the model to detect objects from an input image. The SSD model architecture employs the VGG-16 architecture as the DNN model backbone to extract the features from the input image. However, the fully connected layer of the VGG-16 architecture is not used in SSD. Instead of the fully connected layer, SSD adds a set of convolution layers to generate features at a different scale in a decreasing order of the feature map size. The SSD model uses a set of predefined bounding boxes of different sizes and aspect ratios to detect faces in any input image. The feature layers added at the end of the base VGG-16 network generate the prediction score for each of the bounding boxes per object class in different feature maps.

Multi-Task Cascaded Convolutional Neural Network (MTCNN) [4] is a popular state-of-the-art face detection framework based on convolutional neural networks. This framework combines both the face detection and face alignment task using three cascaded neural networks, Proposal Network (P-Net,) Refine Network (R-Net), and Output Network (O-Net). The first simple network in the pipeline is P-Net, which produces all candidate regions containing facial images in a target image. R-Net is the second network that works on the output of the P-Net, refines the overlapped bounding boxes, and rejects the false selection of the first network. O-Net is the final network of this pipeline, takes the output of R-Net, and produces the face location in detail by localizing five facial landmarks, including left eye, right eye, nose, left mouth corner, and right mouth corner.

Another popular state-of-the-art algorithm we will explore for face detection is YOLOv5 [8]. YOLO: "You Only Look Once" [5–8], means the neural network applies only one forward propagation for each sample. Therefore, the algorithm achieves high speed and accuracy. To detect faces, this algorithm divides input images into a grid and predicts bounding boxes using dimension clusters, so the network predicts four points for each bounding box. After that, for each bounding box in the image, using a multilevel classifier, the algorithm predicts the class of the object bounded by the box. For this class prediction simple logistic classifier is implemented here. The bounding box prediction is made on three scales in the convolution network for any object. Later, these three predictions are merged to get more meaningful semantics and finer-grained image features.

The backbone layer is responsible for extracting all important features from an input image. YOLOv5 employs CSPDarknet, which uses cross-stage partial networks to perform feature extraction [26]. It reduces the network parameter size and reuses the extracted features. This new hybrid network is faster, performs better, and is more powerful than other networks used in previous versions of YOLO [5–8].

3.2 Face Recognition Algorithms

The face recognition trend started long ago in computer vision, basically recognizing a person's face from images based on facial features. Though face recognition is a straightforward and rudimentary task for humans, it has proven to be a complex task for computers, as many factors can affect the resultant accuracy. Nevertheless, many state-of-the-art algorithms are available in face recognition, starting from traditional to deep convolution neural network methods.

Local Binary Pattern Histogram (LBPH) [11,12] is one of the popular and easiest face recognition algorithms based on the local features of the image. This algorithm is a combination of two methods - Local Binary Pattern (LBP) and Histogram of Oriented Gradient (HOG) [10]. LBP is a very powerful visual descriptor for texture classification, and it was first described in 1994. LBP method converts each pixel of an image from grayscale value to binary value using thresholding on the 3×3 neighbors of each pixel. LBPH algorithm populates an intermediate image from the original image to highlight facial features using the LBP method. Now using the concept of the HOG method, LBPH generates a localized histogram of this intermediate image based on binary values of each pixel. To generate a histogram, the binary image is divided into a grid of 8×8 cells and populated histogram of each grid cell separately. During training, the LBPH algorithm creates a histogram for each image from the training dataset. In the recognition phase for any input image, LBPH firstly generates a feature histogram, then compares this histogram with all histograms of the training image dataset, and returns the image ID with the lowest dissimilarity. If the similarity measure is less than some predefined threshold, it is considered an unknown face.

FaceNet [14] is another state-of-the-art face recognition model based on Deep Convolution Neural Network (DCNN). DCNN-based previous face recognition models [15,16] employ a selection layer at the end of the network to classify face images. Nevertheless, these approaches are inefficient because of the bottleneck layer used at the end of the deep network. Moreover, the number of parameters representing face embedding is also huge (1000s of dimensions). But in FaceNet, the network is trained to generate face embedding of 128-D for each face. This embedding is used to measure direct Euclidean distance between different face classes to recognize face identities. Triplet loss [14], a new loss function, is used in FaceNet to train the network, so the distance between similar faces becomes low. In contrast, for a dissimilar face, the network generates higher distances. To calculate triplet loss, two face images are considered with each anchor, one is a positive face image of the same class, and another is a negative face image from a

different class. The target is to adjust the network so that the distance between the anchor and positive face image is smaller, whereas the distance between the anchor and negative face is as greater as possible. After generating 128-D feature embedding of any face image, we can find the class of that face image by using simple k-NN classification.

Deep Convolution Neural Network (DCNN) based face recognition methods use different loss functions to train the network, so that feature embedding has a smaller distance for the same class and a larger distance for different classes. Some DCNNs directly produce the image classification by using a final fully connected layer at the end of the network and the softmax loss function [9,24, 25]. However, some DCNNs learn the embedding rather than the class score using triplet loss [14] like functions. Both of these loss functions have some drawbacks when the number of training classes increases and the number of iterations increases significantly [17]. Additive Angular Margin Loss (ArcFace) [17], a new loss function proposed to improve the distance between classes near the class boundary using the arc-cosine function. The idea of this loss function is that the angular distance between the target network weight and the current feature embedding is calculated using the arc-cosine function and to increase this angular distance in the feature hypersphere, an additive angular margin is added with it. Using this loss function ArcFace optimizes the distance margin and increases the inter-class distance near the boundary, which increases the network power to separate the images of one class from other classes.

4 System Architecture

The architecture of the system consists of several crucial components. To capture face images from all room positions, we have used a PTZ Camera (pan, tilt, and zoom). The PTZ control system controls the movement and focus of the camera during video recording. The PTZ system allows us to schedule how the camera will move when the attendance system is used for face detection and recognition. Based on the prerecorded schedule, the PTZ control system will move the camera to capture all person's faces in the room during the session. The camera sends captured live video to the face detection and recognition module for further processing.

The face image repository contains the face images of all students in a class or persons who will present at a gathering where we want to apply the system. All these face images must be mapped with an identification number resulting from the attendance system after face recognition. The face detection and recognition module is another core component of the attendance system. This module is responsible for detecting all faces available in a frame of the live video captured by the camera. All the detected faces are separated from the frame, and then these faces are compared with the faces available in the repository to find the person's identity in the image. So the face recognition module takes a live video feed from the camera and face images from the repository as input to identify the attendance of all persons available in the camera's frame. The output of the face

detection and recognition module is the bounding box for each face image where the face image is located in the video frame with the identification number and matching score of the detected face with the identified person. The recognized face images are stored in the repository frame-by-frame for further analysis and cross-checking. The identification number of recognized persons is then stored in the attendance database with the time and other related information (Fig. 2).

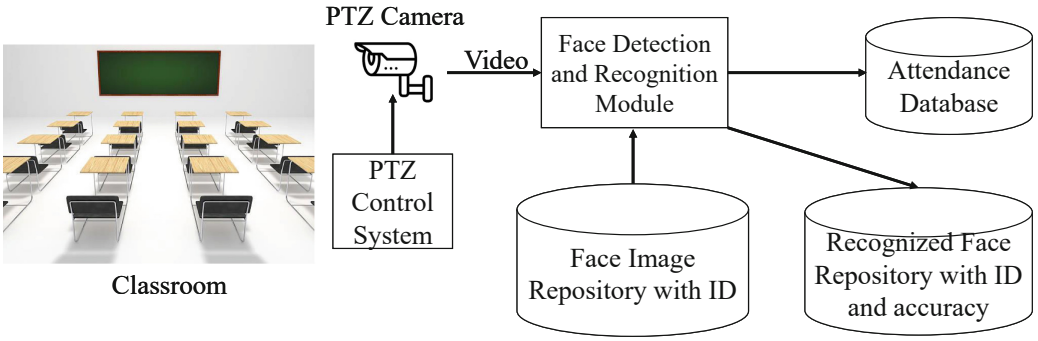


Fig. 2. Architecture of the face recognition based attendance system.

The architecture of the face recognition empowered attendance system can be decomposed into 2 phases - the training and recognition phases. Firstly in the training phase, we have to collect students' information together with sample face images of around 50 nos per student. After taking images, various preprocessing tasks have been conducted based on the face recognition model we want to train on sample images. Then the face recognition model will be trained using preprocessed face images to learn distinct features of individual students' faces so that the algorithm can distinguish one student's face from other students' faces. A critical fact is that the student's sample face images are captured in the same environment as the classroom so that the model learns robust features that will impact the recognition task. The training phase in this architecture is a one-time process, which means it needs to run model training only once for the entire student life. Here we have trained LBPH [11,12], FaceNet [14], and ArcFace [17] models using the same face dataset. The trained features and the respective student's ID are stored in a database for future use in the recognition phase.

The recognition phase is the real-time evaluation in this attendance system architecture and the most crucial step in considering the performance. In this phase, the system captures real-time video of the classroom during the lecture period and provides this video stream as input to the face recognition algorithm. Here the system performs two major tasks - face detection and face recognition. After receiving the input video stream, the face detection algorithm converts the video into separate frames and detects all faces possibly available in each video frame. Here we have applied HaarCascade [1–3], SSD [27], MTCNN [4], and YOLOv5 [8] to compare the face detection performance on the same input. The

detected faces are cropped from the video frame as individual face images and forwarded to already trained face recognition algorithms, LBPH [11,12], FaceNet [14], DeepFace [16], and ArcFace [17]. The face recognition algorithm extracts relevant features like the size of eyes, noses, mouth, jaw, the distance between eyes, the distance between nose and mouth, etc. After finding distinct features, the recognition algorithm measures the similarity between the features stored in the database during the training phase for sample face images of all students. The face recognition algorithm returns the student ID of the most similar student's face from the database with the input face image. This recognized student ID is recorded in the attendance system for that specific lecture period, and for each period, we need to run this recognition process once. The main challenge in this real-time face recognition is the variation of light effect and the scaling effect due to the classroom length and face movement. To verify the usability of state-of-the-art algorithms in this environment, we will examine different face detection and face recognition algorithms in this study (Figs. 3 and 4).

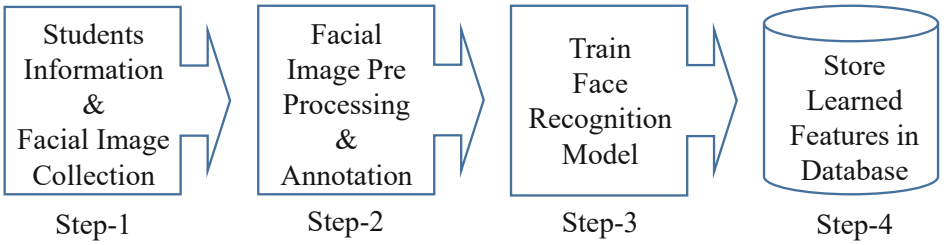


Fig. 3. The Training Phase of Attendance System on collected student's face images.

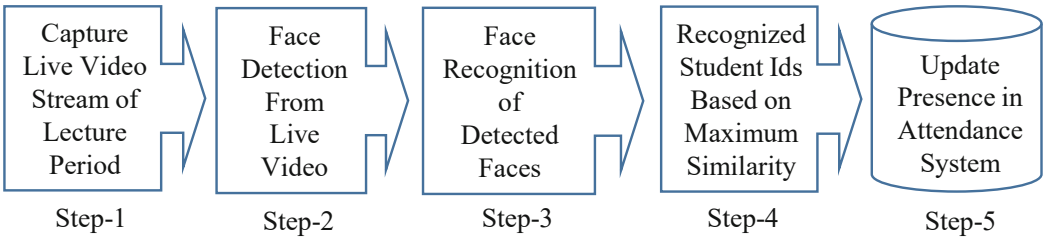


Fig. 4. The Recognition Phase of Attendance System on live video stream during lecture period.

5 Dataset Collection

Many datasets are available to work on face detection and recognition-based projects. These datasets are robust and well recognized where the images are

collected from social media and the internet, and most of these images are captured in outdoor conditions. However, our application is quite different from the general-purpose face recognition projects. We need to detect and recognize faces in an indoor environment from a large gathering of people. Therefore we have created our dataset for this project by collecting face images using this system in a classroom. Here we have collected the name and student ID of each student we want to register in the system. Together with this information, we have captured a number of face images of the respective student in different poses, face angles, and brightness. After capturing the required number of face images, we have to save the information of the registered student in the database (Fig. 5).

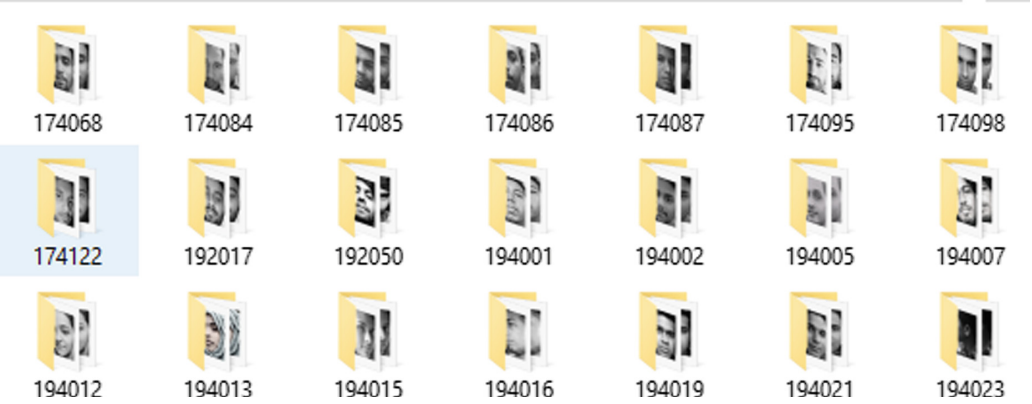


Fig. 5. Face image repository.

The system will create a directory for each registered student in the repository and place the images inside the directory. These images will be used to train the face recognition module during the training phase of the face detection and recognition module.

Here we have collected face images in different positions and face angles for a student as a student will likely not sit in the same position and pose for all the classes. Figure 6 shows the face images of a student we collected to train the face recognition model.

6 Training the System and Recording Attendance

We did not train our model from initial weight values to make the face recognition module more robust and unbiased to our collected dataset. For that, we have used pre-trained modes in the training phase. We have collected pertained models of ArcFace, FaceNet, and DeepFace, and trained on our collected dataset. After reading the entire dataset, the training dataset is divided into two parts, one for training the model and another for validating the model after each training process. Firstly we read each image from the training dataset and apply the face detection model to detect the face from the image, and this process removes the



Fig. 6. Face images in different face angles and position.

unwanted region of the image. Then the detected face is aligned according to the location of the eyes, nose, and lips. Next, we resize the image if the detected face image size is larger than 120×120 pixels. Finally, the preprocessed face images are used to train the face recognition model. We have used the stochastic gradient descent optimizer to optimize the face recognition model with a 0.0005 initial weight decay parameter. After training the face recognition model, the final weight parameters of the deep convolution network are stored in a file. This trained model weight will be used for face recognition during recording attendance.

The system will turn on the camera and start the video recording to record the attendance at a gathering. This real-time video recording is then provided to the face detection module to detect all the faces in each video frame. After detecting the face images, the system performs the resize and alignment operation to make the face image more recognizable. The result of the face detection model is the cropped faces from the video frame. These face images are then applied to the previously trained ArcFace model to recognize the identity of the person's face present in the image. The result of the ArcFace face recognition model is the bounding box of the recognized face in the video frame, the identification number of the recognized face, and the matching accuracy of the input face with the recognized face image. All this information is rendered on the video frame and the computer display so the teacher can check the result. In addition, the identification numbers of the recognized students are also shown in this user interface.

After the face recognition process, the students whose face was recognized successfully are recorded as present in the database with the corresponding date.

The students who are not present in the class and captured video are recorded as absent in the database (Fig. 7).

date	id	course	teacher	status
Filter	Filter	Filter	Filter	Filter
2022-05-25	84040	CSE-2821	Omar	A
2022-05-25	18204012	CSE-2821	Omar	A
2022-05-25	18204030	CSE-2821	Omar	A
2022-05-25	18204031	CSE-2821	Omar	A
2022-05-25	18204040	CSE-2821	Omar	A
2022-05-25	84040	CSE-2821	Omar	A
2022-05-25	84055	CSE-2821	Omar	P
2022-05-25	18204012	CSE-2821	Omar	A
2022-05-25	18204030	CSE-2821	Omar	A
2022-05-25	18204031	CSE-2821	Omar	A

Fig. 7. The attendance record in the database.

7 Experimental Setup and Results

To evaluate face recognition models, we have selected 120 students and taken 50 images of each student in the same classroom environment to create the training image dataset. A PTZ camera has been installed in the classroom from where a full view of the classroom can be captured. As female students attend class wearing hijabs, we have also included images of some students with hijabs. To make the training robust, we have collected images with different face angles, poses, resolutions, and brightness. During preprocessing, we have cropped the face region from each image using a face detection algorithm, and then necessary alignment is done on the cropped image. The final image dataset contains images of 120×120 pixels. Here we have evaluated the face detection models, HaarCascade, SSD, MTCNN, and YOLOv5. We also have trained and tested LBPH, FaceNet, DeepFace, and ArcFace face recognition models for the face recognition phase on our created dataset.

All of these models are trained on the same image dataset. We have trained all models in the same computer with NVIDIA GeForce 930MX GPU. To evaluate these trained models, we have recorded a video of students in the same classroom to analyze the real-time situation. After the face detection process,

Table 1. Accuracy Comparison

Face Detection Model	Detection Accuracy (%)	Face Recognition Model	Recognition Accuracy (%)
HaarCascade	75	LBPH	66.67
SSD	85	FaceNet	78.44
MTCNN	95	DeepFace	86.23
YOLOv5	95	ArcFace	94.74

we have stored the detected faces in a directory, and these images we have used in the face recognition model to measure the model's accuracy. Table 1 shows the face detection accuracy and the face recognition accuracy of different models. The face detection accuracy of MTCNN and YOLOv5 is the same for our experimental setup and higher than HaarCascade and SSD. From the recognition models, the ArcFace model is more accurate than other models considered here. ArcFace model gives better results when used with the YOLOv5 face detection model. Based on this evaluation of the real-life dataset collected in the classroom environment, we have used YOLOv5 as the face detection model and the ArcFace face recognition model in this automated attendance system.

8 Conclusion and Future Works

In this work, we have compared the performance of four face detection models and four face recognition models when applied to the automated attendance system in a real-life environment. The main goal of this experiment is to explore the robustness of different state-of-the-art face detection and recognition models in real-life situations considering various factors. The result shows that the YOLOv5 face detection model and ArcFace face recognition model give accuracy higher than 90%. In contrast, the accuracy of other face recognition models is lower than the ArcFace model. Based on this analysis we have designed an automatic attendance system for mass gathering using the YOLOv5 and ArcFace mosel. In further research, we will try to evaluate other face recognition models using our real-life dataset for the automated attendance system used in the indoor environment. Another research direction is analyzing students' behavior using this system and measuring the quality of education through this behavior analysis.

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